

CHERNIHIV, Ukraine

WMO: 331350

Lat: 51.4439N

Lon: 31.2003E

Elev: 463

StdP: 14.45

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.7	2.4	-8.2	3.6	-1.2	-3.4	4.7	3.7	21.0	30.7	19.0	28.8	5.9	60	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	87.8	68.7	84.6	67.6	81.5	66.2	71.1	83.2	69.5	80.7	68.0	78.6	6.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	101.6	76.4	65.6	96.2	74.6	64.1	91.0	73.0	35.2	83.2	33.8	81.0	32.6	78.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
18.8	16.6	14.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-8.1	92.5	6.3	3.6	-12.6	95.0	-16.3	97.1	-19.9	99.2	-24.4	101.8		
			-8.6	73.7	6.2	2.2	-13.1	75.3	-16.7	76.5	-20.2	77.7	-24.7	79.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	23.5	25.4	34.4	48.6	59.1	65.4	69.0	66.8	57.1	45.6	35.6	26.9	
	DBStd	18.18	10.45	10.59	8.36	7.65	7.79	6.32	5.68	6.17	6.99	7.52	8.57	10.26	
	HDD50	3467	820	688	484	115	15	0	0	0	17	178	435	715	
	HDD65	7111	1285	1108	947	493	214	74	25	50	251	602	882	1180	
	CDD50	2217	0	0	2	73	296	462	589	521	230	41	3	0	
	CDD65	387	0	0	0	1	31	86	149	106	14	0	0	0	
	CDH74	3423	0	0	0	16	291	689	1271	1008	146	2	0	0	
	CDH80	980	0	0	0	2	52	159	398	341	28	0	0	0	
Wind	WSAvg	7.1	7.8	7.9	8.2	7.4	7.0	6.7	6.2	6.1	6.4	6.7	7.5	7.8	
Precipitation	PrecAvg	24.0	1.5	1.6	1.4	1.7	2.5	2.5	2.9	2.3	2.1	1.9	1.9	1.8	
	PrecMax	31.4	2.9	3.2	3.2	3.9	5.2	5.2	5.7	4.9	4.6	6.3	4.0	4.5	
	PrecMin	17.6	0.4	0.7	0.2	0.1	1.0	0.3	0.9	0.2	0.1	0.0	0.1	0.3	
	PrecStd	3.0	0.6	0.7	0.7	0.9	1.0	1.3	1.5	1.1	1.4	1.4	0.9	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	47.1	60.4	75.5	84.9	88.0	91.5	93.6	83.7	71.3	57.3	48.1	
		MCWB	41.3	41.9	48.1	58.0	65.7	67.9	70.6	70.8	63.6	58.5	52.9	46.1	
	2%	DB	40.4	42.6	55.1	70.3	80.8	84.5	87.8	87.9	78.3	65.1	52.7	44.3	
		MCWB	38.2	39.5	45.4	55.2	64.2	66.9	69.3	69.1	62.7	56.8	49.8	42.8	
	5%	DB	37.7	39.5	50.2	65.8	77.1	81.2	84.7	83.5	73.9	60.4	49.4	41.5	
		MCWB	36.1	37.0	42.8	52.9	62.2	66.2	68.5	67.3	60.8	54.3	47.1	40.3	
	10%	DB	35.2	36.9	46.1	61.9	73.3	78.0	81.2	79.5	69.5	56.5	46.7	38.3	
		MCWB	33.8	34.8	40.8	50.8	60.4	64.7	67.1	65.0	58.7	51.9	44.9	36.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.2	43.4	49.9	60.9	68.6	71.2	73.3	72.8	66.5	61.5	53.5	47.0
			MCDB	43.6	46.6	59.1	74.4	80.2	82.6	85.5	86.6	79.4	67.5	55.7	48.1
2%		WB	38.5	39.8	46.3	56.7	66.0	69.1	71.6	70.6	63.9	57.9	50.3	42.9	
		MCDB	39.8	41.7	52.7	66.4	77.6	80.1	83.6	83.8	75.5	63.2	52.2	43.9	
5%		WB	36.2	37.3	43.7	54.2	63.8	67.7	70.0	68.8	61.8	55.0	47.6	40.5	
		MCDB	37.6	39.3	49.2	63.1	74.1	78.6	80.8	80.7	71.2	59.7	49.1	41.5	
10%		WB	34.0	35.2	41.3	52.1	61.8	66.1	68.5	67.0	59.7	52.4	45.0	36.9	
		MCDB	34.9	36.7	45.2	60.4	71.4	75.4	78.3	77.1	67.4	56.0	46.6	37.9	
Mean Daily Temperature Range		5% DB	MDBR	7.9	9.8	12.6	17.1	18.7	17.7	17.9	19.4	17.2	14.0	8.3	7.4
			MCDBR	7.2	11.3	19.7	23.2	23.5	22.4	23.3	26.1	24.6	20.7	12.2	8.6
	MCWBR		6.6	8.7	11.8	11.3	10.3	9.3	8.7	10.3	11.0	12.2	9.3	7.6	
	5% WB	MCDBR	7.0	10.2	18.0	20.4	21.2	20.2	20.6	23.0	21.7	18.7	10.5	8.2	
		MCWBR	6.6	8.3	11.7	10.7	10.2	9.5	8.8	9.9	10.8	12.2	8.6	7.5	
Clear-Sky Solar Irradiance	taub	0.306	0.338	0.369	0.431	0.406	0.394	0.426	0.428	0.396	0.361	0.342	0.309		
	taud	2.170	2.102	2.167	2.124	2.276	2.331	2.258	2.236	2.286	2.350	2.270	2.209		
	Ebn at Noon	215	240	258	255	269	273	261	253	248	234	201	193		
	Edh at Noon	23	33	38	44	40	38	41	39	34	26	22	19		
All-Sky Solar Radiation	RadAvg	259	525	953	1358	1779	1883	1817	1573	1075	586	233	170		
	RadStd	44	89	142	136	133	157	129	112	136	100	36	41		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.12	N/A	N/A	+1.89	+1.35	+1.13	N/A	-328	+175	N/A
(47)	Regional (8 neighbors)	+1.17	N/A	N/A	+1.78	+1.03	+1.01	-212	-308	+182	+83

Nomenclature: See separate page